

Dijkstra Algorithm

Conditions

- It is important to note the following points regarding Dijkstra Algorithm
- Dijkstra algorithm works only for connected graphs.
- Dijkstra algorithm works only for those graphs that do not contain any negative weight edge.

Dijkstra Algorithm

- The actual Dijkstra algorithm does not output the shortest paths.
- It only provides the value or cost of the shortest paths.
- By making minor modifications in the actual algorithm, the shortest paths can be easily obtained.
- Dijkstra algorithm works for directed as well as undirected graphs.

Dijkstra's Algorithm

Dijkstra's Algorithm

Step 1. Set $\lambda(s) = 0$ and for all vertices $v \neq s$, $\lambda(v) = \infty$. Set $T = V$, the vertex set of G . (We will think of T as the set of vertices “uncoloured”.)

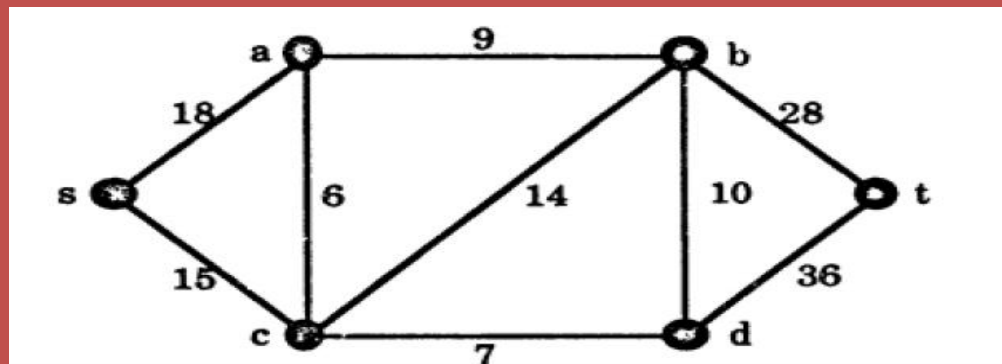
Step 2. Let u be a vertex in T for which $\lambda(u)$ is minimum.

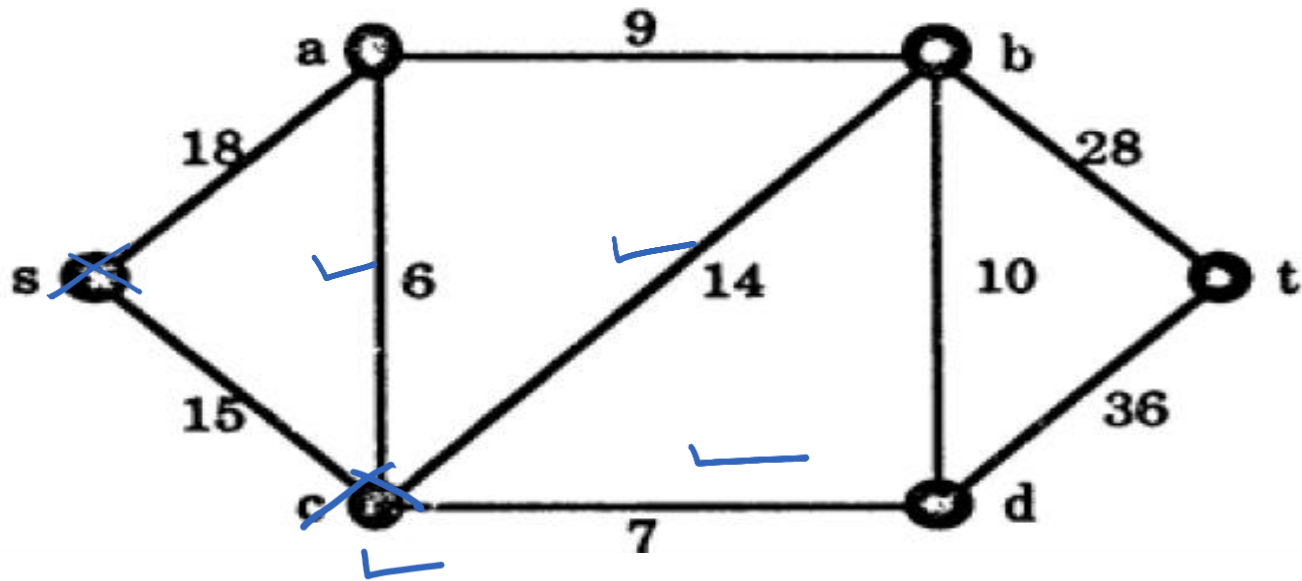
Step 3. If $u = t$, stop.

Step 4. For every edge $e = uv$ incident with u , if $v \in T$ and $\lambda(v) > \lambda(u) + w(e)$ change the value of $\lambda(v)$ to $\lambda(u) + w(e)$ (i.e., given an edge e from an “uncoloured” vertex v to u , change $\lambda(v)$ to $\min(\lambda(v), \lambda(u) + w(e))$).

Step 5. Change T to $T - \{u\}$ and go to Step 2, (i.e., “colour” u and then go back to Step 2 to find an “uncoloured” vertex with minimum label).

Q. Using Dijkstra's algorithm find the shortest path and length of the shortest path from the vertex s to the vertex t of the following weighted graph.





$(s) \xrightarrow{18} a \xrightarrow{9} c \xrightarrow{28} (t) \rightarrow ? \quad ()$

$s \xrightarrow{15} c \xrightarrow{7} d \xrightarrow{36} t = ()$

$s \xrightarrow{18} a \xrightarrow{9} c \xrightarrow{14} b \xrightarrow{28} t = ()$

Step 1

Vertex v	s	a	b	c	d	t
$\gamma(v)$	0	<u>∞</u>	<u>∞</u>	∞	∞	∞
T	s	a	b	c	d	t

Step 2 $U = s$, $\gamma(s)$, minimum value

Step 3 $U = t$ $\square$

Solution:

Step 1. The initial labelling is given by:

vertex v	s	a	b	c	d	t
$\lambda(v)$	0	∞	∞	∞	∞	∞
T	{ s ,	a ,	b ,	c ,	d ,	t }

Step 2. $u = s$ has $\lambda(u)$ a minimum (with value 0).

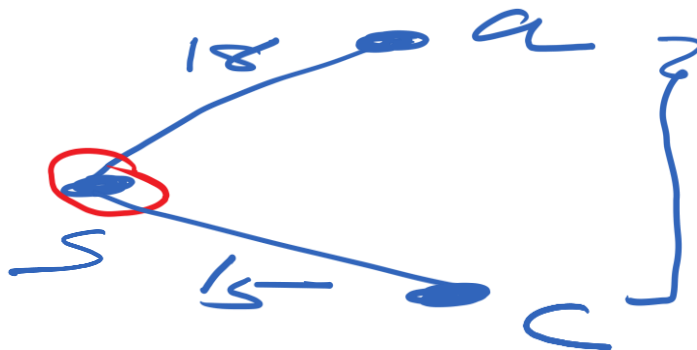
Step 4. There are two edges incident with s , namely $\{s, a\}$ and $\{s, c\}$. Both a and c are in T .

$$\lambda(a) = \min(\lambda(a), \lambda(s) + w(\{s, a\})) = \min(\infty, 0 + 18) = \min(\infty, 18) = 18.$$

$$\lambda(c) = \min(\lambda(c), \lambda(s) + w(\{s, c\})) = \min(\infty, 0 + 15) = \min(\infty, 15) = 15.$$

Step 5. Reset $T = T - \{s\} = \{a, b, c, d, t\}$. Thus, we have

vertex v	s	a	b	c	d	t
$\lambda(v)$	<u>0</u>	<u>18</u>	∞	<u>15</u>	∞	∞
$\checkmark T$	{	$a,$	$b,$	$c,$	$d,$	$t\}$



$$\begin{aligned} \lambda(a) &= \min \{ \lambda(a), \lambda(s) + w(s, a) \} \\ &= \min \{ \infty, 0 + 18 \} \\ &= 18 \\ \lambda(c) &= \min \{ \lambda(c), \lambda(s) + w(s, c) \} \\ &= \min \{ \infty, 0 + 15 \} \\ &= 15 \end{aligned}$$

Vertex v	s	a	b	c	d	t
$\pi(v)$	0	18	20	15	22	29
T	X	a	b	c	d	t

$$U = \underline{\underline{C}} ; \pi(a), \pi(b), \pi(c)$$

$$\begin{aligned} \pi(a) &= \min \{ \pi(a), \pi(c) + w(c, a) \} \\ &= \min \{ \underline{\underline{18}}, 15 + 6 \} = 18 \end{aligned}$$

$$\pi(b) = 29 \quad \underline{\quad}$$

$$\pi(d) = 22 \quad \underline{\quad}$$

$$\lambda(b) = \min(\lambda(b), \lambda(a) + w(\underline{b, a}))$$

Step 2. $u = c$ has $\lambda(u)$ a minimum and u in T (with $\lambda(u) = 15$).

Step 4. There are three edges incident with c , namely $\{c, a\}$, $\{c, b\}$, and $\{c, d\}$. Each of a , b , and d is in T .

$$\lambda(a) = \min(\lambda(a), \lambda(c) + w(\{c, a\})) = \min(18, 15 + 6) = \min(18, 21) = 18.$$

$$\lambda(b) = \min(\lambda(b), \lambda(c) + w(\{c, b\})) = \min(\infty, 15 + 14) = \min(\infty, 29) = 29.$$

$$\lambda(d) = \min(\lambda(d), \lambda(c) + w(\{c, d\})) = \min(\infty, 15 + 7) = \min(\infty, 22) = 22.$$

Step 5. Reset $T = T - \{c\} = \{a, b, d, t\}$. Thus, we have

vertex v	s	a	b	c	d	t
$\lambda(v)$	0	18	29	15	22	∞
T		a	b	c	d	t

Step 2. $u = a$ has $\lambda(u)$ a minimum for u in T (with $\lambda(u) = 18$).

Step 4. There is one edge incident with a , namely $\{a, b\}$ with b is in T .

$$\lambda(b) = \min(\lambda(b), \lambda(a) + w(\{a, b\})) = \min(29, 18 + 9) = \min(29, 27) = 27.$$

Step 5. Reset $T = T - \{a\} = \{b, d, t\}$. Thus, we have

vertex v	s	a	b	c	d	t
$\lambda(v)$	0	18	27	15	22	∞
T			$b,$		$d,$	$t\}$

Step 2. $u = d$ has $\lambda(u)$ minimum for u in T (with $\lambda(u) = 22$).

Step 4. There are two edges incident with d , namely $\{d, b\}$ and $\{d, t\}$ with b, t are in T .

$$\lambda(b) = \min(\lambda(b), \lambda(d) + w(\{d, b\})) = \min(27, 22 + 10) = \min(27, 32) = 27.$$

$$\lambda(t) = \min(\lambda(t), \lambda(d) + w(\{d, t\})) = \min(\infty, 22 + 36) = \min(\infty, 58) = 58.$$

Step 5. Reset $T = T - \{d\} = \{b, t\}$. Thus, we have

vertex v	s	a	b	c	d	t
$\lambda(v)$	0	18	27	15	22	58
T	{		$b,$			$t\}$

$\lambda(b) = 27$

;

$\lambda(t) = 58$

Step 2. $u = b$ has $\lambda(u)$ minimum for u in T (with $\lambda(u) = 27$).

Step 4. There is one edge incident with b , namely $\{b, t\}$ with t in T .

$$\lambda(t) = \min(\lambda(t), \lambda(b) + w(\{b, t\})) = \min(58, 27 + 28) = \min(58, 55) = 55.$$

Step 5. Reset $T = T - \{b\} = \{t\}$. Thus, we have

vertex v	s	a	b	c	d	t
$\lambda(v)$	0	18	27	15	22	55
T	{					t }

Step 2. $u = t$, the only choice.

Step 3. Stop.

When the algorithm stops the values $\lambda(v)$ give the lengths of the shortest paths from the vertex s to each vertex v . Thus the lengths of these paths from s to a, b, c, d, t are 18, 27, 15, 22 and 55 respectively.